# Symmetry as a Design Axis in Variational Quantum Learning

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***Abstract*—The success of variational quantum learning models crucially depends on choosing parametrizations that reflect the structure of the problem at hand. Symmetries provide one of the clearest such structures: whenever transformations of the input leave the desired outcome unchanged, this invariance should be built into the model rather than discovered during training. However, imposing a symmetry does not by itself determine a useful ansatz. Even within the symmetry-preserving space, one still has to decide where the trainable degrees of freedom should be placed. In this work, we study this remaining design freedom in equivariant variational quantum circuits. Building on symmetry-based parameter sharing, we disentangle two architectural choices that are often conflated: how much of the available symmetry should be enforced, and which symmetry-respecting interactions should be made trainable. Using Tic-Tac-Toe as a fully enumerable and structurally transparent test case, we find that relaxing full symmetry to suitable subgroups preserves most of the generalization benefit. By contrast, the dominant performance gains arise from gates that act directly on the decisive motifs of the task. Our results suggest a general principle for quantum ansatz design: symmetry is a powerful way to restrict the search space, but the effectiveness of the model is determined by placing trainable interactions where the structure of the problem actually lives.**

***Index Terms*—quantum machine learning, equivariance, symmetry, inductive bias, variational quantum circuits, parameter sharing, generalization**

## I. Introduction

Symmetry provides a precise form of prior knowledge about a learning task. Consider a target map $y : \mathcal{X} \to \mathcal{Y}$ and a symmetry group $S$ acting on the input space through transformations $V_s : \mathcal{X} \to \mathcal{X}$. The task is invariant under $S$ if

$$y(V_s[x]) = y(x) \qquad \text{for all } x \in \mathcal{X},\ s \in S.$$

In words, a symmetry identifies transformations of the input that should not change the correct prediction. A reflected cat image still depicts a cat; a rotated or mirrored Tic-Tac-Toe board still has the same winner. A model that already respects this structure need not spend data, parameters, and optimization effort learning separate rules for symmetry-related inputs.

This idea is central to geometric deep learning: architectures are designed around the transformations under which a task should be invariant or equivariant. Convolutional networks are the classical example, reusing filters across translated image patches [1]. In quantum machine learning, the analogous program is often described as geometric quantum machine learning (GQML): known group actions are represented on the Hilbert space and used to constrain quantum models by construction [2], [3]. Related work has developed group-invariant quantum models, general equivariant quantum neural networks, permutation-equivariant architectures, graph-equivariant circuits, and symmetry-informed variational ansatze [2]–[5].

Variational quantum learning models (VQLMs) are a natural setting for this idea. They are parameterized quantum circuits trained as statistical models [7], made expressive through data re-uploading [8], but this expressivity is useful only when it is shaped by the structure of the task. Meyer et al. [6] give a concrete blueprint for doing so. If the data embedding satisfies

$$U(V_s[x]) = U_s U(x) U_s^{\dagger},$$

then the data symmetry is represented unitarily on the Hilbert space. Choosing trainable blocks that commute with this representation, and combining them with an invariant initial state and invariant readout, yields a model whose prediction is invariant by architecture rather than by optimization.

We build directly on this construction, but do not treat symmetry as a complete ansatz specification. A known task symmetry has several architectural consequences: it can determine which inputs are equivalent, which gates should share parameters, which observables should be averaged, which subgroups may be sufficient, and which task motifs should receive trainable interactions. Our central question is therefore diagnostic: which of these symmetry-induced structures need to be hard-coded into the circuit, and which can be learned from data once part of the equivariant bias is already present?

Tic-Tac-Toe provides a deliberately transparent benchmark for this question. Tic-Tac-Toe is $D_4$-invariant, finite, and structurally transparent: all legal, reachable boards can be enumerated, and the label-relevant motifs are exactly the three-cell winning lines.

Our starting point is the $D_4$-equivariant Tic-Tac-Toe ansatz of Meyer et al. [6]. In that model, each board field is encoded on one qubit by a Pauli-$X$ rotation with angle $x_i = 2\pi g_i/3$, and the board symmetries act as permutations of the qubits. The trainable circuit respects this geometry by sharing parameters over symmetry orbits: single-qubit rotations are tied across corners, sides, and the center, and directed controlled-$R_Y$ rotations are tied across symmetry-related pairs. This edge-based ansatz already shows the central benefit of equivariance: enforcing the correct board symmetry improves generalization over an unconstrained circuit.

We use this ansatz as a baseline and separate two design choices that are usually entangled. First, we vary how much of the board symmetry is enforced, from full $D_4$ equivariance down to smaller subgroups. This tests whether all symmetry

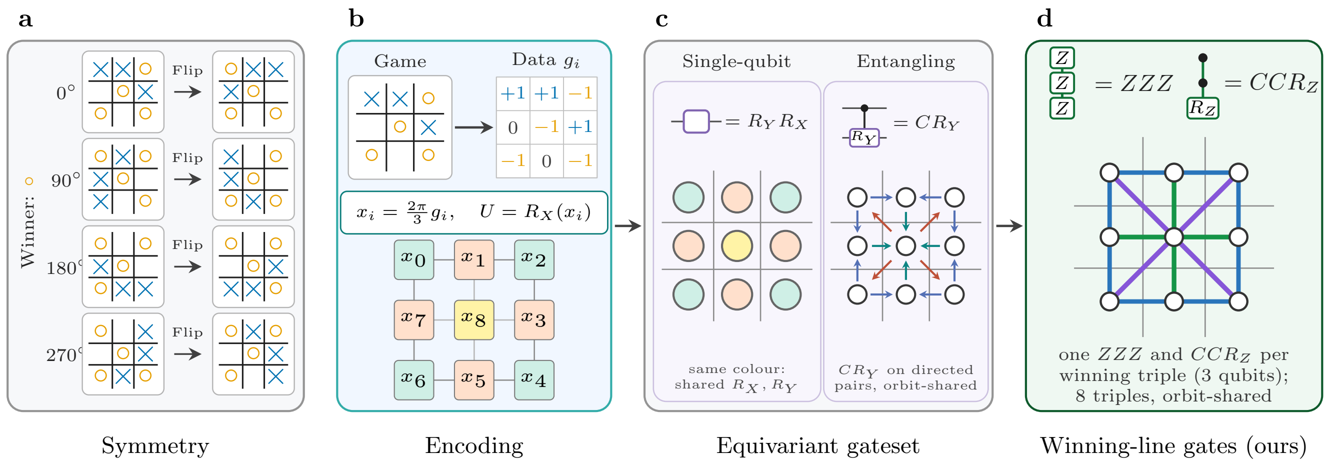


Fig. 1. From task symmetry to a task-aligned equivariant circuit. (a) Tic-Tac-Toe labels are invariant under the $D_4$ rotations and reflections of the board. (b) Board entries $g_i \in \{-1, 0, +1\}$ are encoded as $R_X(x_i)$ rotations with $x_i = \frac{2\pi}{3} g_i$; the chosen indexing exposes the corner, side, and center orbits. (c) The Meyer et al. [6] edge ansatz uses orbit-shared $R_Y R_X$ rotations and directed $CR_Y$ entanglers on neighbouring and center-connected fields. (d) Our extension adds orbit-shared $ZZZ$ and symmetric $CCR_Z$ gates on the eight winning triples, directly matching the label-relevant motifs while preserving equivariance. Panel layout adapted from Meyer et al. [6].

relations have to be imposed explicitly, or whether weaker sharing can preserve most of the inductive bias while leaving the circuit more flexible. Second, we vary which symmetry-respecting interactions the circuit is given to learn. The original edge ansatz uses directed controlled-$R_Y$ gates to couple neighbouring fields, but a Tic-Tac-Toe outcome is decided by complete three-cell lines. We therefore add trainable gates directly on the rows, columns, and diagonals that determine the label, while still sharing their parameters across symmetry orbits.

Our results support a refined view of symmetry in variational quantum learning. Enforcing the full board symmetry is beneficial relative to an unconstrained circuit, but relaxing it to suitable subgroups preserves most of that benefit in this benchmark. The larger improvement comes from the second axis: placing equivariant trainable gates on the label-relevant winning-line motifs. Thus, symmetry should not be treated as a single switch to turn on or off. It defines a constrained design space, and the remaining architectural question is where inside that space the trainable degrees of freedom should live.

## II. Protocol

Each legal, reachable Tic-Tac-Toe board is represented by nine values $g_i \in \{-1, 0, +1\}$ for circle, empty, and cross. The three labels are circle win, draw, and cross win, encoded as $(+1, -1, -1)$, $(-1, +1, -1)$, and $(-1, -1, +1)$. As in Meyer et al. [6], unfinished games are counted as draws. Since the full set contains only 5478 legal, reachable boards, we enumerate it exactly and draw class-balanced train and test splits from this set. The labels are invariant under the eight rotations and reflections of the board, forming the dihedral group $D_4$; this symmetry is illustrated in Fig. 1(a).

We use one qubit per field, as shown in Fig. 1(b). Each value $g_i$ is mapped to the angle $x_i = 2\pi g_i/3$ and encoded by an $R_X$ rotation,

$$U(g) = \bigotimes_{i=0}^{8} R_X\left(\frac{2\pi}{3} g_i\right).$$

The fields are numbered cyclically around the rim, with the center last. This makes the three $D_4$ position orbits explicit: corners $\{0, 2, 4, 6\}$, sides $\{1, 3, 5, 7\}$, and center $\{8\}$.

We separate the imposed symmetry from the choice of gates. Let $H \subseteq D_4$ be the subgroup that we enforce. For $h \in H$, let $V_h$ act on the board and let $P_h$ be the corresponding qubit permutation. The encoding is equivariant: transforming the board before encoding is equivalent to encoding first and then permuting the qubits,

$$U(V_h g) = P_h U(g) P_h^\dagger.$$

A trainable block $W_H(\theta)$ preserves this symmetry if

$$P_h W_H(\theta) P_h^\dagger = W_H(\theta) \qquad \text{for all } h \in H.$$

In practice, we enforce this by sharing parameters across gates whose supports lie in the same $H$-orbit. For directed gates, the order of the support is kept fixed, so control and target roles are not accidentally exchanged. Thus the symmetry fixes the sharing pattern, but not the gate family itself.

The circuit uses repeated data re-uploading layers. Each layer applies the encoding $U(g)$ followed by $p$ repetitions of a trainable block. Unless stated otherwise, we use $L = 3$ layers and $p = 2$ repetitions, keeping depth fixed across all sweeps.

Our baseline is the edge ansatz of Meyer et al., shown in Fig. 1(c). It applies local $R_Y R_X$ rotations and directed controlled-$R_Y$ entanglers on neighbouring rim fields and

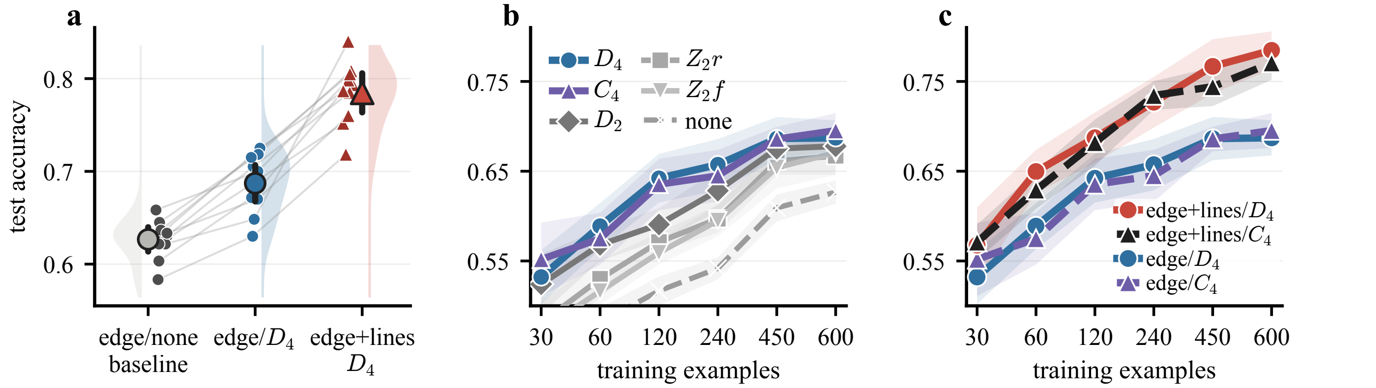


Fig. 2. Main empirical evidence. (a) Full $D_4$ equivariance improves the original edge ansatz over no sharing. (b) The subgroup sweep across training sizes shows that $C_4$ tracks $D_4$ closely. (c) Adding winning-line interactions to the equivariant edge ansatz improves test accuracy while preserving the imposed symmetry.

selected center connections. Symmetry enters only through parameter sharing: gates in the same $H$-orbit share the same parameter or parameter tuple.

The readout averages Pauli-$Z$ expectations over the three position orbits,

$$r_O(g) = \frac{1}{|O|} \sum_{i \in O} \langle Z_i \rangle_g,$$

for corners, sides, and center. These three values are used as class scores. Since the readout orbits are invariant under all subgroups considered here, the readout is compatible with every symmetry sweep.

This gives our first design axis: the amount of symmetry imposed. We keep encoding, edge wiring, gate family, depth, and readout fixed, and vary only the subgroup $H$. With $r$ denoting a $90^\circ$ rotation and $f$ a reflection, we compare $\{e\}$, $\langle r^2 \rangle$, $\langle f \rangle$, $\langle r \rangle = C_4$, $\langle r^2, f \rangle = D_2$, and $\langle r, f \rangle = D_4$. For the edge ansatz at $L = 3, p = 2$, these variants have 204, 108, 126, 60, 78, and 54 trainable parameters, respectively.

The second design axis is the interaction structure. The edge ansatz couples neighbouring fields, but Tic-Tac-Toe labels are decided by complete three-cell winning lines. We therefore add orbit-shared gates on the eight winning triples: three rows, three columns, and two diagonals, as shown in Fig. 1(d). Each triple receives one $ZZZ$ rotation and one $\mathrm{CCR}_Z$ interaction. The resulting edge+lines model keeps the original edge ansatz but adds trainable interactions directly on the label-relevant motifs. At $L = 3, p = 2$, the full-$D_4$ edge model has 54 parameters, while the full-$D_4$ edge+lines model has 90.

## III. Results

This section evaluates the two design axes introduced above: the strength of imposed symmetry and the choice of trainable interactions. We first compare no sharing, partial subgroup sharing, and full $D_4$ sharing with the edge circuit fixed. We then add winning-line interactions and use controls to separate symmetry-aware sharing from mere parameter reduction. All runs use mean-squared error, Adam with learning rate $0.01$ [9], batch size 15, 30 minibatch updates per epoch, 100 epochs, fixed seed blocks, PennyLane simulation [10], and PyTorch optimization [11].

Across these experiments, three trends emerge, summarized in Fig. 2. First, the unconstrained edge ansatz has more parameters but generalizes worse than the $D_4$-equivariant edge model. At 600 training examples, edge/none reaches about $0.63$ test accuracy, while edge/$D_4$ reaches about $0.69$ with a smaller train-test gap. This supports the basic equivariance effect: the correct board symmetry improves generalization by shaping the hypothesis class.

Second, the symmetry effect is not strictly all-or-nothing. Full $D_4$ remains the natural symmetry choice, but $C_4$ rotation sharing tracks it closely across the data grid and is slightly ahead at the smallest training size. We interpret this as evidence that symmetry strength is a tunable architectural bias: partial symmetry can preserve most of the benefit while leaving the circuit less constrained. The random-sharing control in Fig. 3(a) confirms that the gain is not caused by parameter reduction alone; parameter-matched sharing without group-orbit structure performs worse.

The largest improvement comes from changing the interaction structure. Adding winning-line gates to the equivariant edge circuit, as shown in Fig. 2(c), raises the best $D_4$ model from about $0.69$ to roughly $0.80$ test accuracy, with the $C_4$ variant close behind. The edge ansatz can only represent rows, columns, and diagonals indirectly, whereas the line gates act directly on the motifs that determine the label. This is the central result: symmetry helps, but task-aligned equivariant interactions help more.

The controls in Fig. 3 rule out simpler explanations. The effect is not arbitrary parameter reduction, because random sharing underperforms orbit sharing. It is also not just parameter count: edge+lines/$D_4$ has 90 parameters, far fewer than the 204-parameter unshared edge baseline, yet performs better. The ablations show that line channels help on their own, but the strongest model combines edge interactions with both winning-line channels.

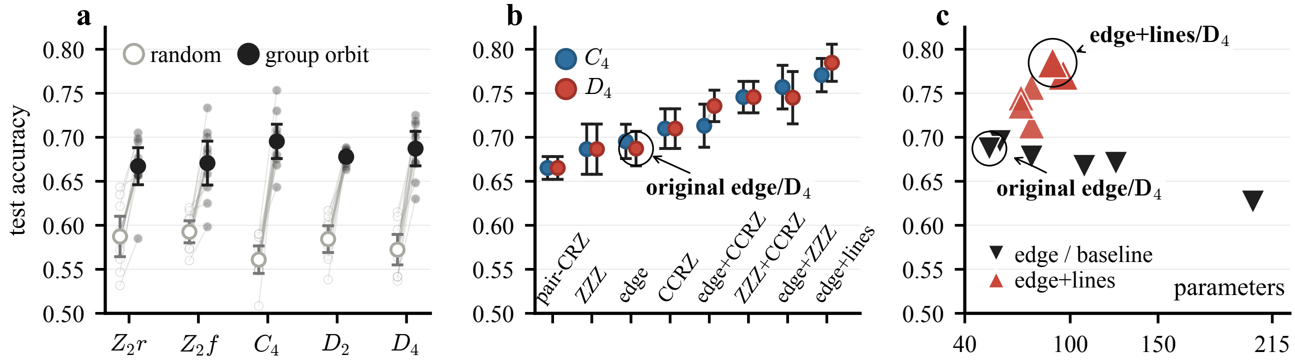


Fig. 3. Controls and ablations. (a) Parameter-matched random sharing underperforms group-orbit sharing. (b) Ablations are anchored by the labelled original edge/$D_4$ ansatz; the strongest model combines edge interactions with both line channels. (c) Accuracy is not explained by parameter count alone.

## IV. Discussion

Our results support a design-oriented view of equivariance in variational quantum learning: symmetry is a powerful inductive bias, but not a complete ansatz specification. It determines how parameters should be shared and how symmetry-related inputs should be treated, yet it does not decide which trainable interactions the circuit should contain. The subgroup sweep shows that equivariance is therefore not a binary choice. Full $D_4$ remains the principled symmetry of Tic-Tac-Toe, but $C_4$ recovers nearly the same generalization behavior, suggesting that partial symmetry can preserve the main inductive bias while avoiding unnecessary constraints. The largest gains instead come from the interaction structure. The original edge ansatz must represent winning lines indirectly, whereas our line-based gates act directly on the three-cell motifs that define the label. Together with the random-sharing and parameter-count controls, this shows that performance is driven not by fewer parameters alone, but by placing trainable degrees of freedom where the task geometry requires them. Tic-Tac-Toe is small and exactly enumerable, but precisely for this reason it isolates a general ansatz-design principle: impose the known symmetry, then choose gates that reach the task-relevant motifs. Future work should test softer relaxations of this recipe, for example shared angles with regularized gate-specific deviations, to interpolate between strict equivariance and fully free circuits.

## Appendix

**Data generation and sampling.** Following Meyer et al. [6], we enumerate the 5478 legal, reachable boards by expanding the game tree and stopping at terminal states; unfinished positions count as draws. Splits are class-balanced; the 600-board test set is fixed and shared across models.

**Reproducibility.** The accompanying anonymous review repository contains the code, checked result tables, figure-generation scripts, and manuscript sources needed to reproduce the reported results, plots, and short paper. This makes the full workflow, from experiment outputs to the final PDF, directly auditable and reproducible.

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